

Lightweight & Unbiased Trust Management System for Resource-Constrained IoT Devices

*Major Project Report submitted in fulfilment of the
requirements for the award of*

Bachelor of Technology
in
Computer Science & Engineering

by

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December, 2025

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Supervisor's Certificate

This is to certify that the project report titled **Lightweight & Unbiased Trust Management System for Resource-Constrained IoT Devices** is being submitted by **Dhruv Khandelwal (2022UCP1130)**, and **Govind Ram Mali (2022UCP1630)** , is a record of original research carried out by them under my supervision and guidance in partial fulfillment of the requirements of the degree of Bachelor of Technology in Computer Science & Engineering. Neither this report nor any part of it has been submitted earlier for any degree or diploma to any institute or university in India or abroad.

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Acknowledgement

We acknowledge the guidance and support of Dr. Arka Prokash Mazumdar, Department of Computer Science & Engineering, MNIT Jaipur. We also thank the lab staff and our colleagues for their valuable feedback during the development of this project.

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Date : December 4, 2025

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Declaration of Originality

We hereby declare that this work entitled **Lightweight & Unbiased Trust Management System for Resource-Constrained IoT Devices** presents our original work carried out as a under graduate student of MNIT Jaipur and, to the best of our knowledge, contains no material previously published or written by another person, nor any material presented by us for the award of any degree or diploma of MNIT Jaipur or any other institution. Any relevant material taken from the the work of others (whether published or unpublished) has been properly acknowledged and referenced in accordance with the institute's guidelines.

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Abstract

IoT networks contain many small, heterogeneous, resource-constrained devices operating in untrusted environments. Existing trust-management approaches rely on large datasets, heavy machine-learning models, manual weight assignment, or central servers — all unsuitable for constrained IoT nodes. This project proposes a lightweight, unbiased, probabilistic trust management system designed to run locally on small devices using minimal interaction data. The system uses only a few simple features (success/failure counts, response-time categories, interaction frequency), computes trust via the Beta reputation model, updates trust event-driven, and detects misbehavior through a deviation-based anomaly test. The approach requires constant small memory, low CPU, and no manual weighting or centralization while naturally capturing uncertainty inherent in small data.

Keywords: *IoT Trust; Beta Reputation Model; Lightweight; Small Data; Event-driven*

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1 Introduction

The Internet of Things (IoT) is a rapidly expanding technological ecosystem consisting of interconnected physical objects such as sensors, actuators, wearables, and gateway devices that collaborate to perform sensing, monitoring, automation, and decision-making tasks. These deployments span a wide range of domains including smart homes, industrial automation, precision agriculture, and healthcare monitoring. Despite their usefulness, many IoT devices operate under severe resource limitations, with constrained processing power, memory, storage, and battery capacity. The heterogeneous nature of IoT environments, combined with their deployment in open, adversarial, or unreliable networks, creates significant challenges for ensuring reliable communication. As devices frequently interact and exchange information, they must be capable of determining whether a peer device can be trusted before accepting data or coordinating operations.

Trust management systems serve as mechanisms for estimating the reliability and expected behavior of devices based on previous interactions and observed performance. A device typically evaluates the correctness, timeliness, consistency, and overall dependability of its peers. However, existing trust management techniques rely heavily on computationally intensive machine-learning models such as LSTM networks or artificial neural networks, depend on manually assigned weights in systems like SMART and fuzzy-logic-based approaches, or require centralized servers for maintaining global trust values. These approaches either introduce bias, require large datasets, or demand computational power and memory unavailable on constrained IoT devices. Consequently, they are unsuitable for real-world deployments involving low-power nodes, which must compute trust in a lightweight, distributed, and efficient manner.

1.1 Problem Statement

The central problem addressed in this work is the design of a trust management system that remains lightweight, unbiased, and computationally feasible for resource-limited IoT devices. The system should operate effectively even when only minimal interaction data are available, eliminating the need for heavy machine-learning models or manually tuned weight parameters. Moreover, it must be able to identify misbehaving or compromised devices in near real time so that corrective actions can be taken promptly within the network.

1.2 Objectives

The primary objective of this project is to create a trust management framework that functions efficiently within the constraints of small IoT nodes. This includes ensuring that trust values can be computed reliably from very limited interaction information while minimizing memory and processing overhead. The system should remain free from subjective weight assignments to maintain fairness and reduce bias in trust evaluations. Additionally, the framework should adopt an event-driven update mechanism to conserve energy by performing calculations only when interactions occur. A further aim is to incorporate a simple yet effective method for detecting deviations or misbehavior using minimal historical data, enabling timely and accurate identification of abnormal device behavior in a dynamic IoT environment.

2 Literature Survey

Trust management in the Internet of Things (IoT) has been studied through several distinct methodological directions, each attempting to evaluate device reliability based on observed behaviour, performance, and contextual factors. The earliest approaches relied heavily on weighted multi-attribute trust models such as SMART, in which multiple indicators—including success ratios, latency, energy usage, and behavioural consistency—are aggregated using pre-defined weights. While these schemes offer flexibility, their effectiveness depends strongly on careful weight assignment, which often requires expert tuning and introduces significant bias. The need for large amounts of interaction data to estimate these attributes accurately makes them impractical for small IoT devices operating under constrained memory and limited communication capabilities.

A second class of approaches uses fuzzy-logic-based trust systems, where linguistic variables (for example, “high reliability,” “moderate delay,” or “low consistency”) are mapped to trust scores using rule sets. Though intuitive, fuzzy systems suffer from rule explosion as the number of input parameters grows. Each additional feature requires new fuzzy rules, increasing memory consumption and computational requirements. These characteristics make fuzzy systems unsuitable for microcontroller-level hardware, where both RAM and processing resources are limited.

Machine-learning-based models, including Support Vector Machines (SVM), logistic regression, and other supervised techniques, have also been employed to classify device trustworthiness. These approaches require large labelled datasets for training and often need to be periodically retrained as device behaviour evolves over time. Their computational and storage demands exceed the capabilities of many IoT nodes, leading to reliance on external servers, which undermines the decentralization and scalability required in distributed IoT environments. Deep learning models, such as Artificial Neural Networks (ANN) and Long Short-Term Memory (LSTM) networks, further extend these challenges. Although they can capture temporal and non-linear behavioural patterns, they demand extensive training data, heavy computation, and large memory footprints, rendering them fundamentally incompatible with resource-limited embedded platforms.

Centralized trust management systems represent another important category, where a central authority maintains and distributes trust scores for all participating devices. Although such systems simplify device-level computation, they introduce considerable communication over-

head, dependency on continuous connectivity, and a clear single point of failure. In dynamic or large-scale IoT networks, the delay in synchronizing trust updates across nodes further reduces the reliability and responsiveness of centralized designs.

Multi-attribute trust models, regardless of their architecture, often require high-cost feature extraction and real-time computation of multiple parameters. These tasks impose additional burdens in terms of power consumption and processing overhead—two resources that IoT devices must conserve.

Table 1: Comparison of Existing IoT Trust Models

Model	Features Used	Limitations	Why Not Suitable for Small IoT Devices
Weighted Multi-attribute (SMART)	Multiple metrics (success rate, delay, energy, consistency) combined using weights	Requires manual weight assignment; sensitive to bias; high pre-tuning needed	Manual weights introduce subjectivity; high computation for multi-attribute scoring
Fuzzy Logic Trust Systems	Rule-based input-output mapping; linguistic variables	Rule explosion as features increase; manual tuning needed; memory heavy	Large rule base consumes RAM; fuzzy inference requires more CPU cycles
Classical ML Models (SVM, Logistic Regression)	Feature vectors, full dataset training, supervised learning	Needs large training datasets; retraining cost high; not adaptive to small data	Training complexity too high; model memory size unsuitable for microcontrollers
Deep Learning Models (ANN, LSTM)	Temporal modeling, multi-layer neural networks	Very heavy computation; large memory footprint; energy intensive	Impossible to run on low-power IoT nodes due to RAM/Flash limits

3 Methodology

This section explains the model, parameters, computation, and anomaly detection with formulas and algorithmic detail.

3.1 Design Principles

The proposed trust management system is guided by several core design principles that ensure suitability for highly constrained IoT environments. First, the computation of trust must remain unbiased, meaning that it should avoid manually assigned weights or heuristics that could introduce subjectivity or designer-induced bias. Instead, the system relies on probability derived directly from observed interactions, allowing trust to emerge naturally from evidence. Second, the model must function effectively even when only very limited historical data are available, since IoT devices may interact infrequently or operate in environments where data collection is sparse.

3.2 Minimal Parameters

For each observed peer, keep:

- α : number of success outcomes observed.
- β : number of failure outcomes observed.
- Response time buckets counters: fast / normal / slow.
- Interaction frequency counter in a sliding window or simple rate.

Table 2: Minimal Parameters Per Peer

Parameter	Description	Type	Cost
α	Success count	Integer	Very low
β	Failure count	Integer	Very low
Response-time buckets	Fast / Normal / Slow counters	3 integers	Low
Interaction frequency	Count or rate of interactions	Integer	Low

3.3 Data Collection

Per interaction record:

- outcome: success (1) or failure (0);
- response-time-bucket $\in \{\text{fast, normal, slow}\}$;
- timestamp.

3.4 Probabilistic Trust Computation: Beta Reputation Model

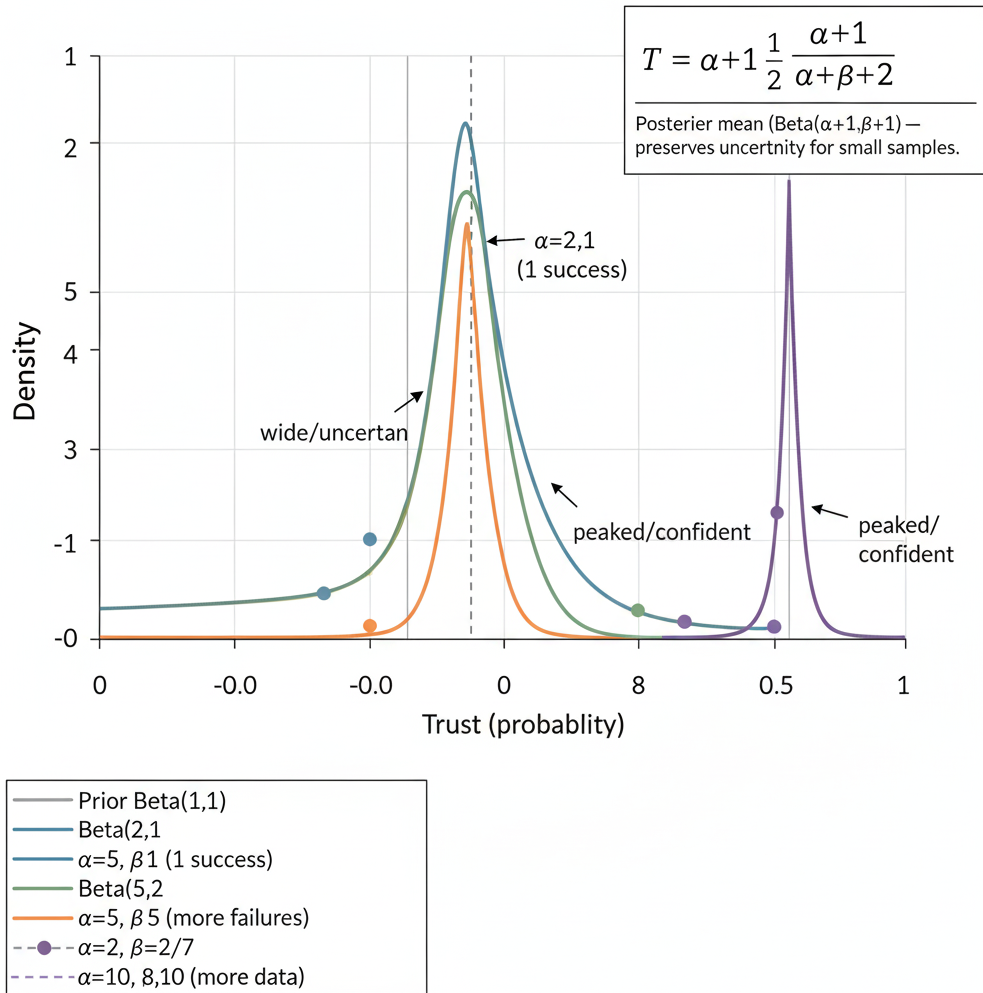


Figure 1: Beta distribution examples illustrating uncertainty with small data.

Compute trust T using:

$$T = \frac{\alpha + 1}{\alpha + \beta + 2} \quad (1)$$

This is the posterior mean for a $\text{Beta}(\alpha + 1, \beta + 1)$ distribution when starting with uniform prior $\text{Beta}(1, 1)$. It preserves uncertainty for small α, β and avoids zero-observation pitfalls.

3.5 Response Time and Frequency Handling

Response-time buckets and frequency are used as lightweight modifiers:

- Slow responses increment a slow counter; a large slow fraction may modestly reduce effective α (or be treated as weaker success evidence).
- Sudden increases/decreases in interaction frequency trigger closer monitoring (e.g., shrink or expand detection window).

Table 3: Response-Time Classification and Trust Impact

Bucket	Response-Time Range	Effect on Trust
Fast	$t \leq 50$ ms	Treated as strong positive evidence (counts fully as success).
Normal	$50 < t \leq 150$ ms	Neutral impact; counted as regular success.
Slow	$t > 150$ ms	Weak success evidence; may slightly reduce effective α .

Rules are intentionally simple to remain computable on microcontrollers.

3.6 Misbehavior Detection: Deviation Test

Maintain a small ring buffer of recent trust values (size N , e.g., 5–10). Compute moving average $\bar{T}$ and standard deviation SD . Flag suspicious if:

$$|T_{\text{new}} - \bar{T}| > k \cdot SD, \quad k \in \{2, 3\}.$$

(2)

3.7 Decision Rule

Given threshold τ (default 0.5) and anomaly flag:

- If $T \geq \tau$ and not suspicious $\Rightarrow$ Trusted.
- Otherwise $\Rightarrow$ Untrusted / Under Monitoring.

τ can be fixed or derived adaptively from local network trust statistics.

Table 4: Trust-Based Decision Rules

Condition	Trust Value (T)	Anomaly Flag	Decision
High trust, normal behavior	$T \geq \tau$	No	Trusted
High trust, abnormal behavior	$T \geq \tau$	Yes	Under Monitoring
Low trust, normal behavior	$T < \tau$	No	Cautious / Limited Access
Low trust, abnormal behavior	$T < \tau$	Yes	Untrusted / Isolate

4 System Workflow

The system architecture illustrated in Figure 2 represents the complete event-driven trust evaluation pipeline followed by an IoT device whenever it interacts with another node in the network. This workflow captures how raw interaction data are collected, transformed into statistical evidence, processed using the Beta reputation model, and finally interpreted to produce a trust-based decision. The design ensures that trust is updated only when necessary, minimizing computational overhead while preserving responsiveness. Each stage in the flowchart corresponds to a lightweight operation tailored for resource-constrained IoT devices, demonstrating how local observations evolve into actionable trust estimates that guide routing, collaboration, and security decisions.

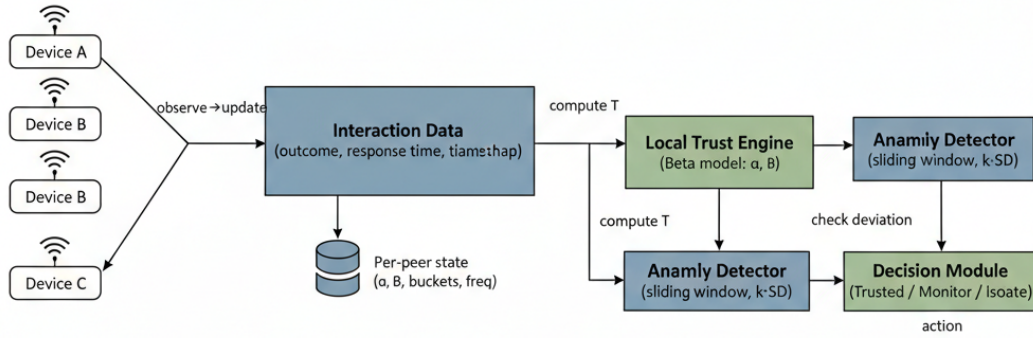


Figure 2: System Architecture

The flow of this event-driven trust management process can be explained in detail as follows:

- **Interaction Initiation:** Whenever device A communicates with device B, an event is triggered. This may involve data exchange, a service request, or a periodic status update. Each interaction becomes an opportunity for device A to evaluate B's behavior.
- **Outcome and Response-Time Collection:** Device A records whether the interaction was successful or failed based on predefined criteria such as packet validity, timeout occurrence, or correctness of data. Simultaneously, the system measures the response time and categorizes it into fast, normal, or slow, providing additional contextual evidence for trust evaluation.
- **Updating Evidence Counters (α and β):** Based on the interaction outcome, the device increments the success counter α or failure counter β . These counters form the core evidence for the Beta reputation model and represent cumulative behavioral history.
- **Trust Computation Using Beta Model:** With updated α and β , the device computes the new trust value T using Equation (1). This probabilistic estimate captures not only reliability but also uncertainty, especially when few observations are available.
- **Sliding Trust Window Update:** The freshly computed trust value is added to a short sliding window that stores recent trust estimates. From this window, the device calculates the moving average $\bar{T}$ and standard deviation SD, enabling temporal analysis of trust behavior.
- **Anomaly Detection:** Using the deviation rule in Equation (3.6), the system determines whether the new trust value deviates significantly from recent patterns. A large deviation may indicate sudden malicious behavior, malfunction, or inconsistent performance.
- **Final Trust-Based Decision:** Based on the computed trust, anomaly status, and optional thresholds, device A decides whether to trust device B. This decision may determine whether data are accepted, routing paths are updated, or the peer is restricted or isolated. The system's lightweight nature ensures decisions are made instantly without interrupting device functionality.

Event-driven Trust Update Flowchart

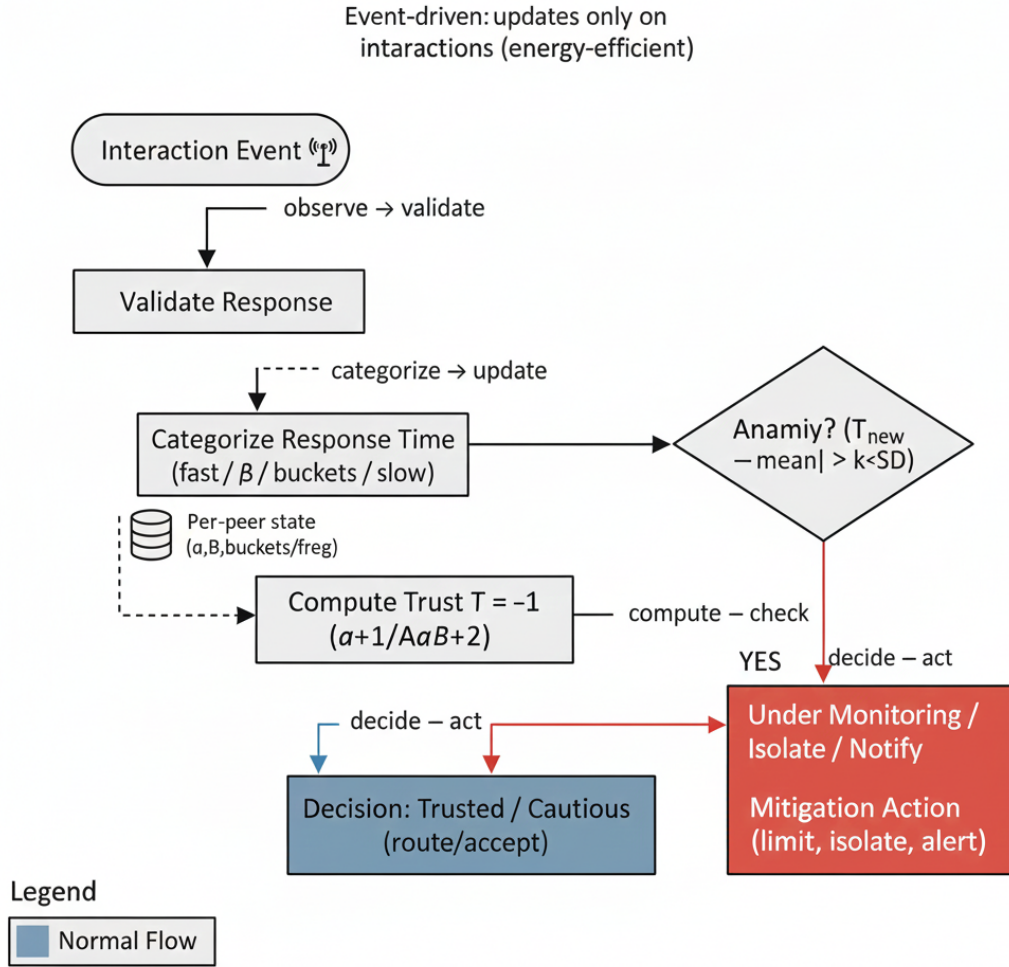


Figure 3: Event-driven trust update flowchart.).

The flow of this event-driven trust management process can be explained in detail as follows:

- **Sliding Trust Window Update:** The freshly computed trust value is added to a short sliding window that stores recent trust estimates. From this window, the device calculates the moving average $\bar{T}$ and standard deviation SD , enabling temporal analysis of trust behavior.
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5 Implementation

The implementation of the proposed trust management system focuses on creating a lightweight, fully local, and computationally efficient framework capable of running on resource-constrained IoT devices. The entire workflow was validated through both dataset-driven analysis and controlled simulations. To evaluate the behaviour of devices under realistic network conditions, we utilized the MQTTset dataset, a publicly available MQTT-based IoT traffic dataset that contains labelled interactions, message delays, anomalies, system faults, and device behaviour traces. This dataset enabled us to reconstruct diverse interaction scenarios, including normal device operation, intermittent failures, and sudden malicious behaviour. By mapping dataset entries to the success–failure outcomes, response-time categories, and interaction frequency metrics required by our model, we were able to test how reliably the Beta reputation model captures trust dynamics in practical settings.

The implementation pipeline consisted of parsing the MQTT dataset, extracting the relevant trust-related attributes, and feeding these into our trust calculation engine. Each interaction record was transformed into the minimal feature set used by our model—namely success/failure outcomes, response-time buckets, and timestamps. These features were then used to increment the α and β counters, apply the Beta trust computation, and update the sliding trust window used for deviation-based anomaly detection. The output of this pipeline included time-series trust values, anomaly flags, and trust-based decisions for every device in the dataset.

In addition to dataset-driven experiments, we implemented a simulation environment to model interactions between ideal (well-behaving) and compromised devices. This simulation allowed us to systematically test the model under controlled conditions, such as gradual trust decay, sudden behavioural shifts, and oscillating misbehavior patterns. The simulator replicates event-driven trust updates and deviation detection exactly as they would occur on an embedded platform, providing an accurate representation of runtime behaviour. The simulated results were used to validate threshold selection, sensitivity to noise, and robustness against false positives.

The core code for dataset processing, trust computation, and simulation was written in Python due to its flexibility and suitability for rapid experimentation. This includes modules for loading the MQTT dataset, computing trust values from interaction logs, performing anomaly detection, and generating visualizations of trust evolution. All implementation code and simulation scripts have been moved to the Appendix section of this report to maintain document

structure while preserving full reproducibility.

In order to maintain a clear separation between conceptual explanation and technical implementation, all source code developed for this project has been included in the Appendix section. The appendix contains the complete Python scripts used for dataset preprocessing, Beta trust computation, sliding-window anomaly detection, and the event-driven simulation environment. Each script is documented with appropriate comments and structured in a way that allows readers to reproduce all experiments presented in this report. By relocating the code to the appendix, the main body of the report remains focused on methodology, analysis, and interpretation, while still ensuring full transparency and replicability of the implemented system.

6 Experiments and Results

6.1 Suggested Evaluation Plan

1. **Simulation:** run the Python script with scenarios: always-good, always-bad, intermittent failures, and sudden compromise.

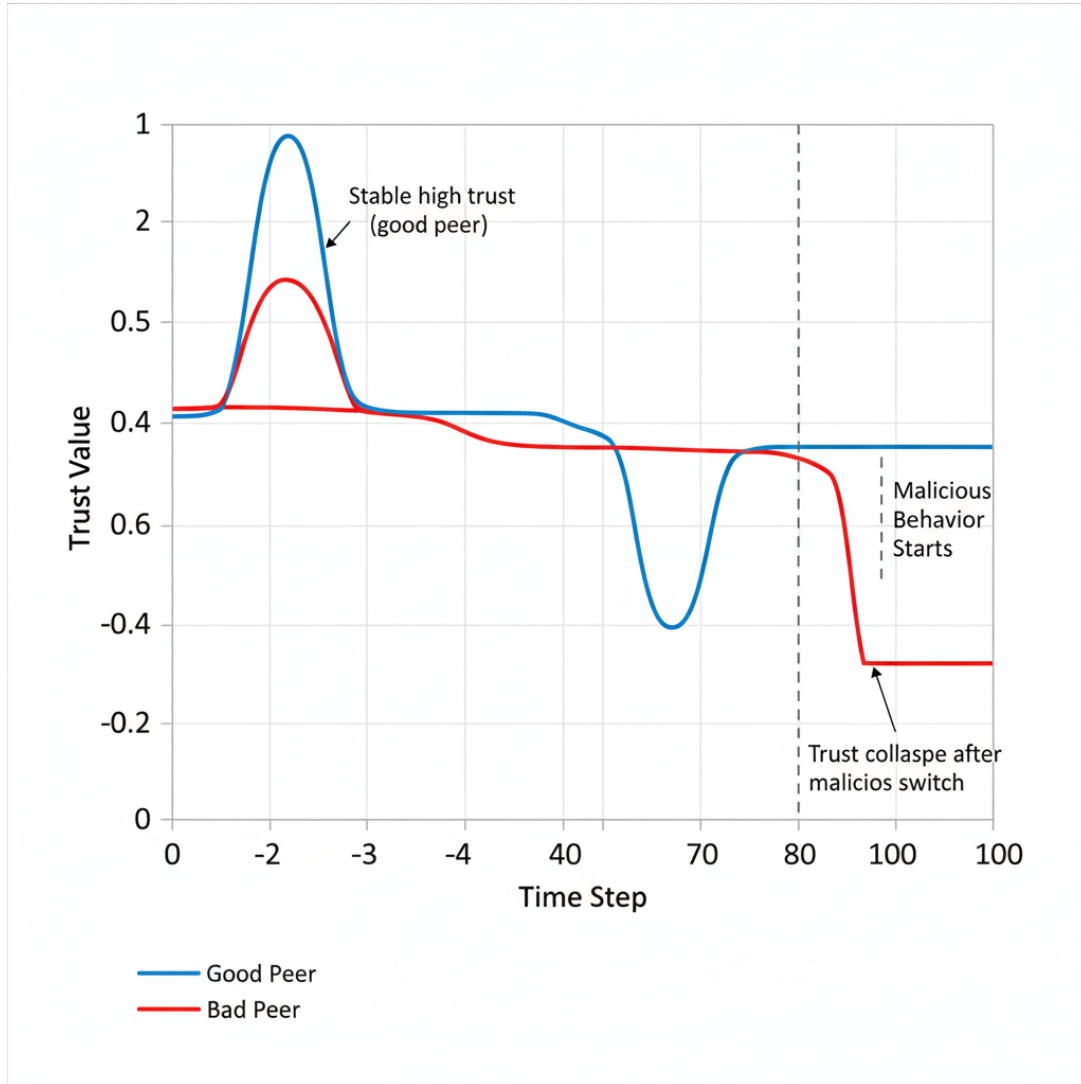


Figure 4: Trust evolution of a good peer and a bad peer..

2. **Testbed:** deploy the embedded C module on devices such as ESP32 or Cortex-M boards. Use a small gateway to orchestrate interactions.
3. **Metrics:** detection accuracy (TPR/FPR), detection time, memory usage, CPU cycles, energy consumption, and communication overhead.
4. **Baselines:** implement a naive weighted model, fuzzy rules, and centralized reputation

system for comparison.

6.2 Expected Results (Hypothesis)

- The Beta-model enables meaningful trust estimates even with very few interactions.
- Event-driven updating reduces CPU usage and energy compared to periodic polling.
- Deviation detection flags sudden behavioral changes quickly with low memory footprint.
- False positive rate is controllable by tuning k and window size.

7 Conclusion and Future Scope

This work presents a small-device, small-data, unbiased trust management system for IoT. The Beta reputation model provides probabilistic trust estimates that are computationally light and unbiased (no manual weights), while deviation-based detection supplies fast misbehavior alerts with minimal storage. The system is suitable for microcontroller-class devices and supports event-driven energy-efficient operation.

7.1 Future Work

Future extensions of this research can focus on enhancing the scalability and robustness of the proposed trust management framework. One promising direction is the incorporation of lightweight collaborative trust propagation mechanisms, such as gossip-based protocols, which would allow devices to share partial trust information without introducing excessive communication overhead. This would enable the network to react more quickly to compromised nodes while still respecting the resource limitations of IoT hardware. Another important extension is the development of adaptive thresholding methods that rely on privacy-preserving network statistics. Such methods would allow each device to dynamically adjust trust and anomaly thresholds according to observed network conditions, improving both detection accuracy and resilience to fluctuating behaviour patterns.

Further research may explore the integration of the trust model across diverse IoT hardware platforms, accompanied by detailed energy profiling to assess real-world performance and identify optimization opportunities. Addressing sophisticated threats such as collusion among malicious nodes or deliberate identity spoofing remains an open challenge, requiring more advanced detection strategies that do not significantly increase computational cost. Additionally, combining the trust framework with lightweight attestation techniques and cryptographic primitives could provide stronger security guarantees, helping authenticate device identity and integrity alongside behavioural trust. These directions collectively offer a pathway toward building a more secure, adaptive, and energy-efficient trust ecosystem for future IoT deployments.

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Appendix A: Python Simulation and Trust Computation Code

This appendix contains the complete source code used during the implementation and evaluation of the proposed trust management system. The scripts included here support dataset preprocessing, Beta reputation trust computation, sliding-window anomaly detection, and simulation of device behaviour under different interaction patterns. These code artifacts allow full replication of the experiments and can serve as a foundation for extending the system to additional datasets or hardware platforms.

Python Simulation and Trust Computation Code

```
1 # Paste your complete simulate_trust.py code here
2
3 """
4 Simple_simulator_for_Beta-based_trust_model_with_deviation_detection
5 .
6 Run:_python_simulate_trust.py
7 """
8
9 import random
10
11 import statistics
12
13 SCALE = 1.0
14 K = 2
15 TAU = 0.5
```

```

15 class Peer:
16     def __init__(self):
17         self.alpha = 0
18         self.beta = 0
19         self.trust_window = []
20         self.win_size = 7
21
22     def observe(self, success):
23         if success:
24             self.alpha += 1
25         else:
26             self.beta += 1
27         t = (self.alpha + 1) / (self.alpha + self.beta + 2)
28         self.trust_window.append(t)
29         if len(self.trust_window) > self.win_size:
30             self.trust_window.pop(0)
31         return t
32
33     def check_anomaly(self, t):
34         if len(self.trust_window) < 2:
35             return False
36         mean = statistics.mean(self.trust_window)
37         sd = statistics.pstdev(self.trust_window)
38         if sd == 0:
39             return False
40         return abs(t - mean) > K * sd
41
42     def simulate():
43         good_peer = Peer()
44         bad_peer = Peer()
45
46         steps = 100
47         bad_switch = 60
48         results = {'good': [], 'bad': []}

```

```

49
50 for i in range(steps):
51     t_g = good_peer.observe(random.random() < 0.95)
52     suspicious_g = good_peer.check_anomaly(t_g)
53     results['good'].append((t_g, suspicious_g))
54
55     if i < bad_switch:
56         s = random.random() < 0.95
57     else:
58         s = random.random() < 0.2
59
60     t_b = bad_peer.observe(s)
61     suspicious_b = bad_peer.check_anomaly(t_b)
62     results['bad'].append((t_b, suspicious_b))
63
64 for name in ['good', 'bad']:
65     arr = results[name]
66     trusted_count = sum(1 for t,s in arr if (t >= TAU and not s)
67                          )
68     print(f"{name}:_final_trust={arr[-1][0]:.3f},_trusted_steps
69           _={trusted_count}/{steps}")
70
71 try:
72     import matplotlib.pyplot as plt
73     ts_g = [x[0] for x in results['good']]
74     ts_b = [x[0] for x in results['bad']]
75     plt.plot(ts_g, label='good_peer_trust')
76     plt.plot(ts_b, label='bad_peer_trust')
77     plt.axvline(bad_switch, color='gray', linestyle='--', label=
78                 'bad_becomes_malicious')
79     plt.legend()
80     plt.xlabel('time-step')
81     plt.ylabel('trust')
82     plt.grid(True)

```

```
80     plt.show()
81 except Exception as e:
82     print("matplotlib_not_available,_skipping_plots:", e)
```

Listing 1: Python Code for Trust Simulation and Anomaly Detection